(b) Solve the equation $\sin^2\theta + 2\cos^2\theta = 4\sin\theta + 3$ for $0° < \theta < 360°$. [5]

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9 The equation of a curve is $y = 4 + 5x + 6x^2 - 3x^3$.

(a) Find the set of values of x for which y decreases as x increases. [4]

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(b) It is given that $y = 9x + k$ is a tangent to the curve.

Find the value of the constant k. [4]

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10 An arithmetic progression has first term 5 and common difference d, where $d > 0$. The second, fifth and eleventh terms of the arithmetic progression, in that order, are the first three terms of a geometric progression.

(a) Find the value of d. [3]

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(b) The sum of the first 77 terms of the arithmetic progression is denoted by S_{77}. The sum of the first 10 terms of the geometric progression is denoted by G_{10}.

Find the value of $S_{77} - G_{10}$. [5]

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11 The function f is defined by $\mathrm{f}(x) = 3 + 6x - 2x^2$ for $x \in \mathbb{R}$.

(a) Express $\mathrm{f}(x)$ in the form $a - b(x - c)^2$, where a, b and c are constants, and state the range of f. [3]

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(b) The graph of $y = \mathrm{f}(x)$ is transformed to the graph of $y = \mathrm{h}(x)$ by a reflection in one of the axes followed by a translation. It is given that the graph of $y = \mathrm{h}(x)$ has a minimum point at the origin.

Give details of the reflection and translation involved. [2]

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The function g is defined by $g(x) = 3 + 6x - 2x^2$ for $x \leqslant 0$.

(c) Sketch the graph of $y = g(x)$ and explain why g is a one-one function. You are **not** required to find the coordinates of any intersections with the axes. [2]

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(d) Sketch the graph of $y = g^{-1}(x)$ on your diagram in **(c)**, and find an expression for $g^{-1}(x)$. You should label the two graphs in your diagram appropriately and show any relevant mirror line. [4]

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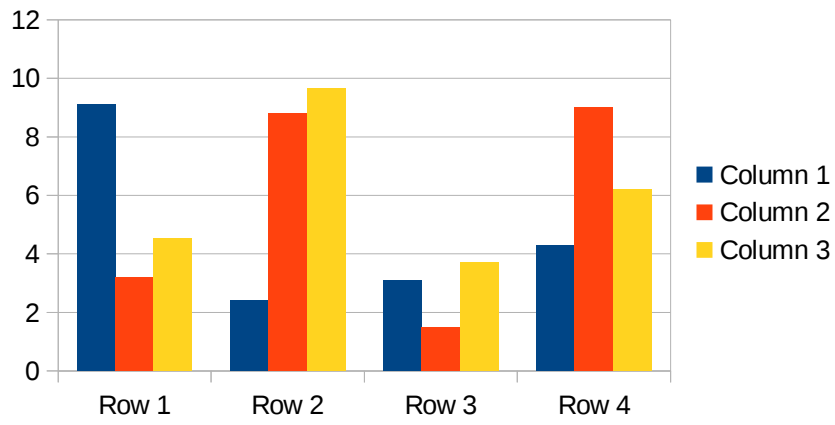
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Translators

Translators

→ Actual Programming code.

- Software's used to convert Source code into object code.
  - Machine Code
  - Ready for execution
  - Understood by CPU
  - Binary Form

1. Compiler
2. Interpreter
3. Java Virtual Machines

Compiler

1. Source Code is provided to compiler
   - Initially no data is provided.
2. Code is analyzed.
3. First Line of code is READ, converted into Intermediate Code
   - Error Free.
4. If a Line having an Error, Error is recorded and we move to the next line until whole code is completed.
5. At end of code, Contain Errors, they are reported — Error Report Generated.
6. If code having no errors, an Executable File is generated. Whole code is converted in one go.

Advantages

- Always generate an executable file
- Executable files cannot be altered since no source code is available.
- Mostly they are Error Free.
- Only Executable file shared, easy to distribute — No source code.
- Executable file can run on different Machines
- Conversion is Fast, since whole code in go.

Disadvantages

- If .exe file is not tested properly, can crash memory.
- Since the whole code is converted in one go, memory can CRASH.
- Finding Errors becomes complex

Interpreters

1. Source code along data is provided
2. Code is analyzed.
3. First Line of code is Read and Executed, if no Errors.
4. If a Line contain Error, Execution is halted
5. Same steps are performed until whole code is converted.

Advantages

1. Finding and Removing Errors is Easy
2. Cannot crash memory.
3. If source code is purchased, interpreter comes along with code.

Disadvantages

- It takes Longer time for execution
- To share code, source code is provided, alterations can be done.
- To execute code, always need interpreter.
- Code running on one machine cannot be interpreted on another.

JAVA VIRTUAL Machines

- Programs are partially compiled (BYTECODE) and partially interpreted
- Hybrid approach and offer portability due to byte code format.
- Platform independent.

- Partially compiled programs can be used on different platforms as they are interpreted when RUN
- Code is optimised as machine code is generated at Run Time.
- Source code does not need to recompiled, more efficient to RUN.

IDE --- Integrated Development Environment

- Softwares in which we perform coding.
- Programs are written and tested by IDEs
- Source Code Editor
- Compiler and Interpreter
- Debugging
- Auto documentation.

Pretty Print

- Displaying or printing well set of formatted source code.
- Easier to read and understand.
- Text colours are changed, underline as well.

Debugging :

- Finding Logical Errors by Running the code and Tracking the variables in variable window watch.

Break Points

- A deliberate pause in the execution of a program during testing so that we can observe values of variables.

Auto documentor

- Provide information about different function
- How to implement certain functions.

Report Window / Watch Window

- Separate window, in Runtime that shows the contents of variables updating during execution of program.

WRITE CODE

- Editor
- Pretty Print
- Expand and collapse into different blocks
- Dynamic syntax check
- Auto formatting
- Context sensitive prompts, help to complete statements

Testing Code

- Stepping - Single
- Break points
- Debugger
- Report Windows